Quantifier clauses for substitution of closed terms

The Open Logic Project contributors

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Open Logic Project unit file sources/olp-first-order-logic-syntax-and-semantics/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: \iftag prob branches are eaten by the preamble tag-macro fix, rendering the complete author branch in both items; the three cited facts (prop:ext-formulas, prop:sat-quant, prop:sentence-sat-true) are carried in-source as Exercise proofs - referenced here as the author's own known facts, stubs built.

Editorial note. The label prop:quant-terms names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own prop environment in the unit source.

Allowed standard prerequisites (editorial).

- Extensionality under substitution for formulas. Same unit, \ollabel{prop:ext-formulas} (source line 931). Core step in both items: Sat(M, subst(A,t',x))[s] iff Sat(M,A)[s with x := value of t'].
- Quantifier clause for variable assignments. Same unit, \ollabel{prop:sat-quant} (source line 662). Converts the closed-term instance to the assignment-wise quantifier clauses.
- Sentences are assignment-independent. Same unit, \ollabel{prop:sentence-sat-true} (source line 649). prvAll final step: A(t) is a sentence, so satisfaction drops the assignment.

Source reference. Original proposition prop-sat-quant: sources/olp-first-order-logic-syntax-and-semantics/pinned-original-source.tex, source lines 662–662. Prerequisite pointer invoked at proof line(s) 2305; it names a standard fact of the same unit that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original proposition prop-sentence-sat-true: sources/olp-first-order-logic-syntax-and-semantics/pinned-original-source.tex, source lines 649–649. Prerequisite pointer invoked at proof line(s) 2313; it names a standard fact

of the same unit that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original proposition prop-ext-formulas: sources/olp-first-order-logic-syntax-and-semantics/pinned-original-source.tex, source lines 931–931. Prerequisite pointer invoked at proof line(s) 2304; it names a standard fact of the same unit that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Proposition 0.1. *Let $\mathfrak{M}$ be structure, and $A(x)$ formula with one free variable x, and t a closed term. Then:*

1. $A(t) \models \exists x\, A(x)$
2. $\forall x\, A(x) \models A(t)$

Proof. 1. Exercise.

2. Exercise.

□